FinAI

Advanced AI Based Personal Finance & Investment Analysis Tool

Revision 3 — Technical Documentation

Market-adjusted Stock Fair Value: fundamentals · technicals · news buzz

Sugamanchi Naveen · 2022WB86112

M.Tech Computing Systems & Infrastructure (WIMS) · BITS Pilani WILP

Supervisor: Soham Das · Additional Examiner: Smrutirekha Behera

October 2026

1. Summary of Revision 3

R3 adds a market-adjusted fair value for each stock. The R2 valuation engine produced several fundamental fair values (DCF, P/E, EV/EBITDA, DDM) and an explainable score, but did not reflect current market conditions. R3 adds two new engines — a news buzz & sentiment engine and a fair value engine — and combines them so that one number answers "what is this stock worth right now?" while every component stays visible.

Area	R2	R3
Valuation	Per-method fair values + blended upside	Market-adjusted fair value + corridor + verdict
Technicals	Indicator values displayed	Indicators converted into a bounded confluence signal
News	Not implemented	Lexicon sentiment, recency decay, buzz level, evidence terms
UI	Chart, indicators, valuation table	Fair Value panel: gauge, waterfall, signal chips, news feed
API	/analysis, /valuation	+ /fair-value; /analysis includes fair_value & news
Launch	START/STOP-FINAI.bat	START/STOP-FINAI-R3.cmd (stops R2 automatically)

2. Architecture changes

The layering rules from docs/01 are unchanged: routes hold no business logic, and the analysis engines are pure functions with no I/O apart from reading fixture files. The new modules sit next to the R2 engines:

```
backend/app/market/   indicators.py   (R2)  SMA, EMA, RSI, MACD, Bollinger,
drawdown, volatility   valuation.py    (R2)  DCF, P/E, PEG, EV/EBITDA, DDM,
sensitivity, scenarios   scoring.py      (R2)  explainable 6-factor stock score
sentiment.py    (R3)  headline scoring, recency-weighted aggregate, buzz
fair_value.py   (R3)  technical confluence + market-adjusted fair value   service.py
orchestration: one analysis payload   fixtures/news/  (R3)  illustrative sample
headlines per symbol
```

Request flow: GET /market/{symbol}/analysis → provider prices & fundamentals → indicators → valuation (intrinsic anchor) → sentiment.aggregate(news) → fair_value.compute_fair_value → JSON.

3. Fair Value methodology

```
fair_value = intrinsic_anchor × (1 + technical_adj) × (1 + sentiment_adj)
```

3.1 Intrinsic anchor

The arithmetic mean of the available fundamental fair values (DCF, P/E, EV/EBITDA, DDM) computed by the R2 valuation engine under the current assumptions. If no method is available, fair value is not produced (null) rather than estimated.

3.2 Technical confluence (bounded ±8 %)

Four signals, each mapped into [-1, 1]; the confluence score is their mean, and technical_adj = score × 0.08.

Signal	Mapping	Reading
Trend	0.7 × clip((price/SMA50 − 1) × 10) ± 0.3 for SMA20 vs SMA50	uptrend / sideways / downtrend
RSI(14)	(RSI − 50)/40 in 30–70; above 70 turns negative (overbought); below 30 positive (oversold)	overbought / neutral / oversold
MACD	clip(histogram / price × 200)	bullish / bearish
Bollinger	clip((0.5 − band position) × 1.2) — near lower band = room to rise	% of band

3.3 News-buzz tilt (bounded ±6 %)

sentiment_adj = sentiment_score × 0.06 × buzz multiplier, where buzz multiplier is 1.0 (high, ≥5 articles in 7 days), 0.8 (elevated, ≥3) or 0.6 (normal). With no news, the tilt is 0 and the news block is marked not_evaluated.

3.4 Corridor and verdict

Corridor half-width = 0.5 × annualised volatility, clamped to 5–25 %. Verdict: Undervalued if price < corridor low; Overvalued if price > corridor high; otherwise Fairly valued, with a mild-upside / mild-downside hint when the gap exceeds ±5 %.

3.5 Why the tilts are bounded

- Fundamentals remain the anchor; market conditions can move the value by at most about 14 % combined.
- This keeps the estimate stable and defensible, and stops one noisy headline or a short rally from dominating.
- Every component (anchor, each tilt, each signal, each headline) is returned by the API and shown in the UI.

4. News buzz & sentiment engine

- Finance-tuned lexicon of weighted words and phrases (e.g. "raises guidance" +1.2, "cuts guidance" −1.2, "downgrade" −1.0).
- Phrases are matched first and removed so words are not double-counted; a preceding negator (not, no, never, without) flips polarity.
- Normalisation: score = x / √(x² + 4), giving a value in (−1, 1). Labels: positive > 0.15, negative < −0.15, otherwise neutral.
- Aggregate: exponentially recency-weighted mean (weight = 0.5^(age/7 days)).
- Evidence: each headline returns its matched terms so the reading can be checked.

4.1 Data and limitations

The bundled headlines in fixtures/news are illustrative sample data written for demonstration and testing. They are not real news and are labelled as such in the files. A live news API can be connected behind load_news() without changing the engine.

Lexicon methods miss context. For example, "attrition declines" is good news for an IT company but scores negative because both words are in the negative list. Experiment 2 (lexicon vs TF-IDF + linear SVM vs FinBERT, measured by accuracy and F1 on a labelled set) is planned to quantify and address this.

5. API reference (new/changed)

Endpoint	Description
GET /api/v1/market/{symbol}/fair-value	fair_value (value, corridor, upside, verdict, components, technical signals, method) and news (sentiment, buzz, scored articles)
GET /api/v1/market/{symbol}/analysis	Unchanged fields plus fair_value and news

Both require a bearer token and return 404 for unknown symbols, consistent with the R2 market endpoints.

6. User interface

frontend/src/components/FairValuePanel.tsx is rendered at the top of the Market page:

- Fair value figure, corridor and upside vs current price, with a coloured verdict badge.
- Gauge showing the current-price marker and fair-value marker relative to the corridor.
- Waterfall cards: intrinsic anchor → technical tilt → news-buzz tilt.
- Signal chips for trend, RSI, MACD and Bollinger with their values.
- News Buzz feed: overall sentiment, positive/negative/neutral counts, each headline with its score and matched terms.

7. Testing

backend/tests/test_fair_value.py adds 8 tests: headline polarity, negation, score bounds and evidence, recency weighting, not_evaluated without news, adjustment caps (±8 % / ±6 %), verdict boundaries, and the endpoint (200 and 404). Run with: docker run --rm -v "%cd%\backend:/app" -w /app python:3.11-slim sh -c "pip install -r requirements.txt && pytest -q".

8. Running R3

- Double-click START-FINAI-R3.cmd. It starts Docker Desktop if needed, stops the R2 stack (same ports), builds and starts R3, seeds demo data and opens http://localhost:8080/app.
- Log in with demo@example.com / Demo#12345 and open Market from the sidebar.
- Double-click STOP-FINAI-R3.cmd to stop. Data is kept; docker compose -p finai-r3 down -v wipes it.

9. Limitations and next steps

- Prices, fundamentals and news are recorded/illustrative samples for two symbols; live providers are the next step.
- Tilt bounds (8 % / 6 %) and the corridor rule are design choices, not fitted parameters. Back-testing the verdicts against later prices is planned for Phase 3.
- Sentiment is lexicon-based; Experiment 2 will compare learned models.

- Phase 3: portfolio risk and expense forecasting. Phase 4: RAG research assistant, hardening and deployment.

FinAI is an educational and analytical tool. Its outputs are not investment advice.